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International A-Level

Physics (9630)
PH03 Fields and their consequences
Report on the Examination

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REPORT ON THE EXAMINATION: INTERNATIONAL A-LEVEL PHYSICS (9630) PH03 – JUNE 2025

QUESTION 01

1.1

Most students were able to use Fleming's left hand rule and realised that the current had to be moving along the wire towards the axis of rotation label. Only a few had the current in a direction not aligned with the wire. Students should be aware that when they are asked to draw an arrow to indicate a direction at a point, then the arrow should pass through the point.

1.2

This required students to calculate the moment of a couple. This was very accessible, although a minority of students doubled the force and used the $\mathbf{5.0}$ cm giving an answer that was too large by a factor of 2.

Some students jumped straight to using a formula that is not on the formulae sheet with no evidence of where it came from. Students should always start from the formula in the formulae booklet as, in determine and show that questions, failure to do this can lose marks even if the answer is correct. In this case, using formulae which used the area did gain full marks, if the answer was correct, but in other questions this may not be the case.

1.3

This question proved accessible. Most students appreciated that the induced emf $\boldsymbol{\varepsilon}$ could be calculated from $\boldsymbol{Blv}$, although some calculated it from first principles. However, some did not use the correct length, instead using lengths from **Figure 1**.

QUESTION 02

2.1

Most students were able to determine the time taken using $\boldsymbol{A = A_0 e^{-\lambda t}}$, although some used the half-life with $\boldsymbol{A = A_0 0.5^{-\frac{t}{t_{0.5}}}}$ and $\boldsymbol{t_{0.5} = \frac{\ln 2}{\lambda}}$.

2.2

This was again accessible with most students able to determine the mass using the radioactive decay formula and the molar mass.

2.3

This question was more challenging. While most students realised that the random nature of radioactive decay meant that whether a nucleus would decay in a given time interval was not predictable, only a few were able to state that the activity could be determined (because $\boldsymbol{N}$ is very large). Many students tried to hedge their bets and wrote about not being able to do so accurately, or there being fluctuations, and so did not gain the mark.

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QUESTION 03

3.1

While a majority of students could determine the maximum frictional force, some did not:

- write down the equation
- substitute the values
- quote the answer to one more significant figure than the approximate value given in the question

as required by 'show that' questions. These students did not get the mark.

3.2

Students had to label the forces acting on a bike and rider travelling around a banked track. This proved challenging to many students. Although the majority of the students correctly identified the direction of the normal contact force, weight and resultant force, most did not identify the *location* of these forces. The weight should be acting at the centre of mass and the normal contact force should be shown acting at the point of contact between the wheel and the track. The resultant force just had to be drawn starting at any point on the rider or bike. The most common answer was to draw all the forces acting at the centre of mass.

3.3

Most students were able to calculate $\boldsymbol{F_2}$; however the question required the students to explicitly compare the magnitude of $\boldsymbol{F_1}$ with the magnitude of $\boldsymbol{F_2}$ and some students did not do this and could only achieve 2 marks.

3.4

The extended writing in this question proved a challenge for most students. While most students were able to show an understanding of the situation, few were able to give enough detail. The idea that the centripetal force in **Figure 4** had a contribution from both the friction and the reaction force was commonly seen. However, not all students explained that only the friction contributed to the centripetal force in **Figure 3**. Only a few were able to clearly articulate that, in **Figure 4**, this achieved a larger centripetal force than the maximum friction and therefore allowed a higher speed than in **Figure 3**.

A commonly expressed idea that was not accepted was that the higher centripetal force caused a higher speed, mixing up the causality of the situation.

QUESTION 04

4.1

This question asked students to define gravitational potential. Most students were able to gain at least some marks here. Common mistakes seen included: referring to energy instead of work done, missing out 'per unit mass' or getting the direction of movement the wrong way round.

Using the equation $\boldsymbol{V = -\frac{GM}{r}}$, even with the terms defined, did not gain the marks as this does not apply to situations beyond a single spherical mass.

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4.2

Students were provided with data for a binary star and the total potential between them and asked to calculate the mass of one of the stars. This was accessible to most students.

4.3

Students were then asked to determine the maximum value of the gravitational field strength from a gravitational potential–distance graph. Students were asked to annotate **Figure 6** to support their calculation. Students were expected to draw a tangent at the steepest point of the graph and determine the gradient of the tangent from a suitably large range of values. Although some students chose a small range of data and therefore did not gain full marks, this was accessible to most students.

A small range of students attempted to use calculus to solve this problem. This is never required on A-level Physics and while some marks could be gained by this approach, the graph also had to be annotated to support the calculation. Unsupported answers did not gain full marks.

4.4

While most students could describe the variation of potential with distance as shown in the graph, many did not state that the gravitational field strength started at 0 (as the line was initially horizontal). Some went beyond the range of the graph and wrote about what happened as the distance approached infinity. This was not penalised this time but, in future, if asked to describe a relationship shown by a graph, students should restrict themselves to what is shown in the graph.

4.5

Students were asked to determine the minimum point at which the variation of $\boldsymbol{g_T}$ with $\boldsymbol{x}$ followed an inverse square law. Most students recognised this as the point at which the curved line became straight. However, many did not gain full marks because they did not explain their method or label their graph appropriately. It was expected that students would draw a straight line on their graph and read off the point where the curve diverged from the straight line.

The most common mistakes were students not taking account of the $\mathbf{ln}(\boldsymbol{x} \,/\, \mathbf{10^{12}}\ \mathrm{m})$ and either not including the $\mathbf{10^{12}}$ or not using $\boldsymbol{e}^{\text{read off}}$ to find $\boldsymbol{x}$.

QUESTION 05

5.1

Most students were able to calculate the potential difference corresponding to a uniform electric field. However, the mixture of significant figures for the data in the question did cause problems for some. When there is a mixture of significant figures for the data, then the minimum number should be used, so in this question the correct number of significant figures is 2.

5.2

Most students were able to calculate the acceleration of the proton, due to the electric field.

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5.3

Students had to explain why the acceleration of the protons was zero inside a hollow metal cylinder in a particle accelerator. Most students found the first two marking points accessible. Most mentioned either no electric field in the cylinder or no potential difference across it or the potential being constant inside the cylinder for the first mark, with no force being the most common way of gaining the second mark. However, very few thought to comment on the significance of the vacuum; there were no collisions with gas particles.

5.4

While most students found the reason for the length of **C2** being longer than the length of **C1**, most did not provide enough detail in their answer. A reference to acceleration was not accepted for protons travelling faster in **C2**, as acceleration means the rate of change of velocity and can refer to slowing down or a change of direction as well as increasing speed. Many students did not quote $\boldsymbol{s = vt}$ or refer to the same time when discussing why the faster speed led to a longer tube.

QUESTION 06

6.1

Some students mixed up the charging and discharging equations for a capacitor, but most were able to calculate that the pd across a charging capacitor was 0.63 after a time equal to one time constant.

6.2

Students had to determine resistance of a resistor from a current-time graph for a charging and discharging capacitor. They were expected to read the $\boldsymbol{y}$ –intercept and use it with the emf to get $\boldsymbol{R = \frac{6}{y\text{–intercept}}}$. Most students gained full marks.

6.3

Students then had to use the same graph to determine the time constant and capacitance of the capacitor. A variety of approaches was possible in this question. The key was to find the time constant and then divide by their answer to **06.2**. However, it was not possible to directly read-off the time constant as the position of the switch was changed before this time was reached.

The most common approaches were:

- read two points from the graph and use the equation $\boldsymbol{I = I_0 e^{-\frac{t}{T}}}$
- read-off the half life and use $\boldsymbol{T = \frac{t_{0.5}}{\ln 2}}$.

Other approaches were possible and any valid route scored full marks.

6.4

Students were then asked to explain the shape of the graph. As is usual on the international specification, students found the extended writing question harder. In each section one bullet point was enough to partially address an area, while three were required to fully address the area.

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Most students were able to describe the changes in the voltmeter reading between $t = 0$ and $t = 7.0\ s$, but only some were able to fully explain the relationship between the meter readings or the changes after $t = 7.0\ s$.

The relationship between the meter readings was most easily expressed using the equation $I = \frac{\varepsilon - V}{R}$ with a comment that as V increases, I must decrease. However, other approaches were possible and were achieved by a small number of students. Most answers had the right concepts but missed out stages, e.g. emf = voltmeter reading + voltage across resistor, or current is the voltage across the resistor divided by the resistance of the resistor. The idea that current and voltage were inversely proportional was rejected as explained by the equation.

The changes after $t = 7.0\ s$ intentionally required more detail, so that only the highest achieving scored full marks. The most commonly made points were that the current was negative due to the direction of the current changing and that it was a discharging circuit. Only the best answers went further than this and wrote about both voltage and current decreasing exponentially or linked the voltage across the capacitor to the voltage across the resistor.

Very few gave answers which discussed the effect of the charge build-up on the capacitor affecting the force pushing electrons onto or off of the capacitor. This was not required for full marks, but was rewarded when seen.

QUESTION 07

Students found the practical aspects of this question the most challenging on the paper. This question asked about required practical 4, a simple pendulum.

7.1

Students had to label the length of the pendulum with a double-headed arrow. Many students did not draw the arrow to either the centre of mass of the bob or to the top of the string. Precision is required in these drawings. Those who did not draw a double-headed arrow as requested did not gain the mark.

7.2

They then had to draw the fiducial marker in the correct position. Many students did not gain the mark because they did not draw the fiducial marker close to the bob. Ideally the card should be (partly) behind the bob and either coming just below or above the bob. For the second mark students had to identify the centre as the place where the bob was moving the fastest or spending the least time. Answers which referred to the decreasing amplitude were not considered enough.

7.3

Students had to explain two techniques to minimise the uncertainty in the measurement of the period. This was the most accessible part of the question where most students were able to score. However, students should make it clear what they mean by repeats. Some answers were unclear about whether the a single period was measured 10 times, or whether there was one measurement of 10 periods.

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7.4

Students had to determine the mistake made to produce some data with an incorrect value for $\boldsymbol{g}$, and then determine the value for $\boldsymbol{g}$ consistent with the data provided. While many students had the right approach to the maths behind this question, many attempted to use the sketch graph rather than the value given. As there was no grid on the graph, it was provided for illustrative purposes only and students were expected to use the value of $\mathbf{39.1}\ \mathrm{m\ s^{-1}}$, rather than estimate the point (1,1). Students who did this lost 1 mark.

As a 'determine' question, students had to show their use of the numbers in the question, rather than just quote the value from the formulae booklet as 9.8(1).

SECTION B

Most of the multiple choice questions were accessible, with all questions except **9** and **17** having greater than two-thirds of the students selecting the correct answer.

Question **17** proved very challenging, with only around a quarter correctly choosing answer **C**. Most (around two thirds) correctly calculated the angle between the plane of the coil and the magnetic field from $\boldsymbol{\sin\theta' = \frac{\phi}{BAN}}$ and chose **B**. They did not realise that $\boldsymbol{\theta}$ as shown in the diagram was the angle between the magnetic field and the plane of the coil and was therefore $\boldsymbol{90 - \theta'}$.

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